

Self evaluation

Criteria	Met?(Y/N)	Comments
The application should include a way for the user to test out their knowledge practically. E.g. simulation	Y	The application enables users to apply the theoretical driving knowledge in practical simulations across a variety of map types including private roads, cities and smaller tracks where each map targets different driving skills and emulates different environments to target skills such as speed control, awareness and rule compliance.
The program should offer environment(s) that simulate real world dynamics	Y	The application includes multiple environments with varying conditions such as visibility, terrain, weather and grip. These factors directly effect user behavior on vehicle handling and braking supporting experimental learning.
The program should include realistic physics and elements (i.e. lights, skid marks etc.)	Y	Realistic driving physics are implemented including acceleration, traction loss, braking behavior, engine ignition and rev meters. Visual elements such as skid marks and headlights further enhance immersion.
The program should include an area for users to test their knowledge	Y	A dedicated test function allows users to complete quizzes after learning or simulation sessions reinforcing knowledge through structured assessments with the program providing feedback for the user.
The users should be given the option of topics they want to cover in said tests	Y	Users can select from three theory topics including road laws, signs and driving rules allowing focused revision based on individual weaknesses.
The program should include progressive difficulty that adjusts difficulty	Y	The application includes five stages of progressive difficulty where advancement is determined by user accuracy. This ensures that content difficulty scales appropriately with user ability
Users should be able to have a choice in customizing their experience and learning. E.g different topic focuses	Y	Users can choose from five learning topics, enabling personalized learning paths and increased engagement
The application offers separated modules for road signs, driving laws and rules	Y	The application is modular separating content into driving laws, rules, scenarios and road signs improving maintainability and usability
The application includes a working sign in/up page	Y	A fully functional logging and sign-up system is implemented using MongoDB, enabling persistent user accounts and progress storage.

refer to appendix C for the Client Evaluation

The primary technical bottleneck in the current implementation is that the quiz logic, difficulty progression and topic selection are handled using tightly coupled conditional statements within single functions. This design negatively impacts its scalability as implementing new topics, levels or types of questions would require modifying existing conditional logic across multiple sections of the code. The maintainability is also effected as changes in one area increase the risk of introducing errors in unrelated logic. To implement this successfully i can refactor using modular design or data-driven architecture where the quiz and progression system could be refactored by separating the quiz logic into independent modules and storing the questions, difficulty thresholds and progression rules in structured data formats such as JSON or database collections. Finally i could use object-oriented approaches instead of nested conditionals.

To improve the simulation component Rule based scoring systems with event-driven evaluation could be implemented such as a penalty system that penalizes users for running red lights. The absence of a scoring or safety rating system reduces user feedback and limits the educational effectiveness of the simulation. This also affects scalability as behavioral evaluation is currently not extensible to more complex driving scenarios. To implement this however Introducing an event-driven system where driving actions (speeding, collisions, signal violations) trigger updates to a safety rating algorithm that aggregates these events over time.

Word count: 451